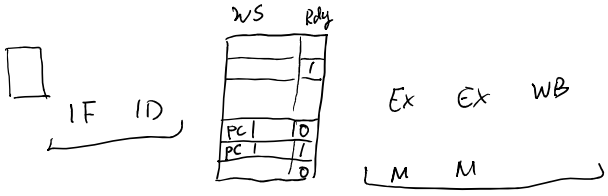
Warp Scheduling and Roofline Model Discussion

Tuesday, February 24, 2026 4:36 PM

1. Scheduling Policy (Warp Scheduling)

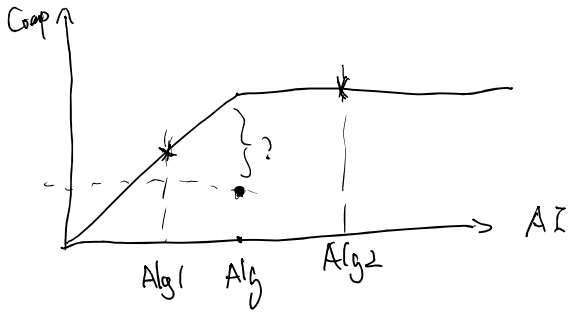


(a) Round Robin Policy
- Fair among warps

(b) Greedy-then-Oldest (GTO)
- Prioritize warps having cache hits

Memory working set
one thread → (A) kB
one warp → 32×A kB
K active warp per SM → K × 32×A kB
L1 $

2. Roofline Model



vector add

BW, FP both underutilized.

- Limited # of threads
- Short-lived threads → kernel fusion
  kernel invocation overhead
- Instruction overhead
- Limited ILP → dependent memory accesses
  pointer chasing